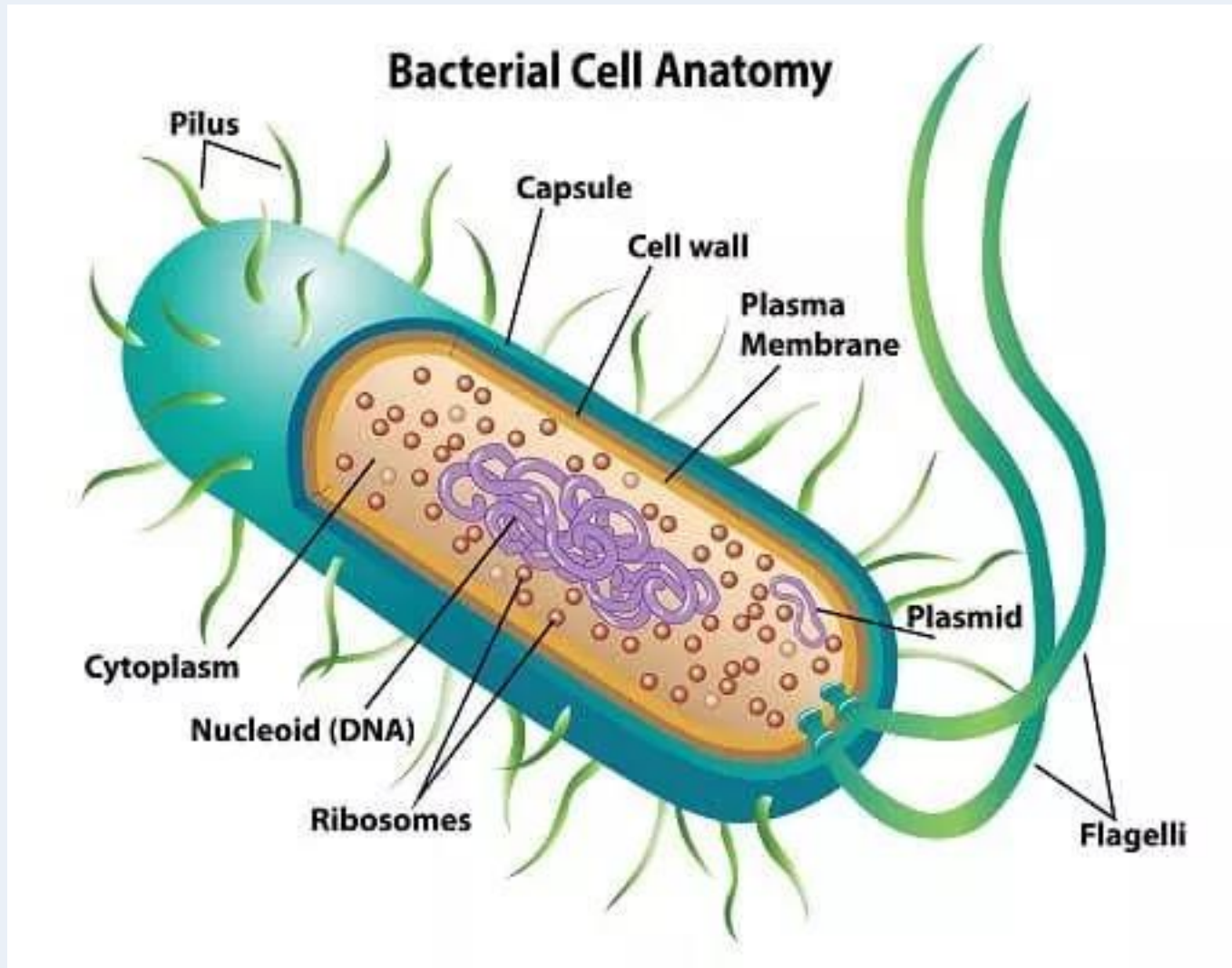




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BACTERIAL CELL STRUCTURE



CELL ENVELOPE :-

- GLYCOCALYX
- CELL WALL
- CELL MEMBRANE

- GLYCOCALYX:-GELATINOUS LAYER COMPOSED OF POLYSACCHARIDE OR POLYPEPTIDE
- GLYCOCALYX IF LOOSE = THEN IT IS KNOWN AS SLIME LAYER
IF THICK = THEN IT IS KNOWN AS CAPSULE LAYER
- FUNCTION OF GLYCOCALYX :- PROTECTION FROM PHAGOCYTOSIS
PREVENT FROM DESICATION

CELL WALL :-

- HARD AND RIGID
- CHEMICALLY COMPOSED OF PEPTIDOGLYCAN ALSO CALLED AS MURIEN
 - PEPTIDOGLYCAN

PEPTIDE

GLYCAN

PEPTIDE = CHAIN OF 4 AMINO ACIDS HELD BY PEPTIDE LINKAGE

GLYCAN = COMPOSED OF N-acetyl glucosamine AND N-acetyl muramic acid HELD BY β 1-4 LINKAGE

THICKNESS OF PEPTIDOGLYCAN:-

FOR GRAM + BACTERIA -- 20-80nm {70-80%}

FOR GRAM - BACTERIA -- 8-12 nm {10-20%}

LIPID LAYER IN CELL WALL OF GRAM + BACTERIA IS LESS

GRAM – BACTERIA IS MORE

➤ FUNCTION:- TO PROVIDE PROTECTION AND DEFINITE SHAPE

➤ CELL MEMBRANE:- IT IS COMPOSED OF LIPID AND PROTEIN {LIPOPROTENACEOUS} AND SELECTIVELY PERMEABLE IN NATURE.

➤ MESOSOME :- INFOLDINGS OF CELL MEMBRANE.

IT IS OF 2 TYPES:-

1. SEPTAL MESOSOMES:- ATTACHED WITH DNA,HELPS IN REPLICATION AND BINARY FISSION
2. LATERAL MESOSOMES:- NOT ATTACHED WITH DNA, HELPS IN RESPIRATION AND SECRETION

GENOPHORE:-

- ALSO CALLED AS NUCLEOID OR NAKED NUCLEUS NOT BOUNDED BY MEMBRANE.
- GENETIC MATERIAL DNA i.e. DOUBLE STRANDED , CIRCULAR.
- ALSO CALLED BACTERIAL CHROMOSOMES.

PLASMID:-

- EXTRA CHROMOSOMAL UNIT i.e. CIRCULAR, DOUBLE STRANDED DNA , SELF REPLICATE.
- PRESENT TIME IT IS USED AS CLONING VECTOR OR VEHICLE DNA IN BIOTECHNOLOGY.
- IT CARRIES FOREIGN DNA AND FORM RECOMBINANT DNA OR CHIMERIC DNA.
- COMMONLY USED PLASMID IS pBR322 CLONING VECTOR.

TYPES OF PLASMID:-

- F-factor PLASMID FOR FERTILITY
- R-factor PLASMID FOR RESISTANCE
- Col-factor PLASMID FOR COLICINOGENY

EPISOME:- PLASMID INTEGRATED WITH BACTERIAL DNA CALLED EPISOME.

RIBOSOME:-

- 70s TYPE CONSISTS OF 2 SUBUNITS
SMALLER i.e. 30S
LARGER i.e. 50S
- CHEMICALLY COMPOSED OF rRNA + PROTEIN
- SMALLER SUB UNIT -rRNA =16S type
Protein=21 type
- LARGER SUB UNIT- rRNA = 23S, 5.8S, 5S
Protein = 34 type

MANY RIBOSOMES HELD BY mRNA CHAINS ARE CALLED AS POLYSOMES.

FLAGELLA :-

- SINGLE STRANDED, 3-12 μ LONG, FILAMENTOUS APPENDAGES.
- CHEMICALLY COMPOSED OF FLAGELLIN PROTEIN.
- DIVIDED INTO 3 DISTINCT REGION :-
 1. BASAL BODY
 2. HOOK
 3. FILAMENT
- FLAGELLA HELPS IN LOCOMOTION.

FIMBRIAE AND PILI :-

- BOTH ARE SIMILAR IN STRUCTURE BUT PILI IS LONGER THAN FIMBRIAE.
- CHEMICALLY BOTH ARE COMPOSED OF PILIN PROTEIN.
- PILI IS ALSO CALLED SEX PILI, SINGLE IS RESPONSIBLE FOR FORMATION OF CONJUGATION BRIDGE DURING CONJUGATION.
- FIMBRIAE PRESENT ALL OVER BODY SURFACE HELPS IN ATTACHMENT WITH SUBSTRATUM AND HOST CELLS.

Thanks